Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Question 6

Part A

$Al_{(s)} \rightarrow Al^{3+}_{(aq)} + 3e^-$

Part B

$3Zn^{2+} + 2Al \rightarrow 3Zn + 2Al^{3+}$

Part C

If $6e^-$ transferred:

mass Zn gained: $\left(\frac{6 \text{ mol } e^-}{1}\right)\left(\frac{3 \text{ mol Zn}}{6 \text{ mol } e^-}\right)\left(\frac{65.38 \text{ g}}{1 \text{ mol Zn}}\right) = 196.14 \text{ g Zn}$

mass Al lost: $\left(\frac{6 \text{ mol } e^-}{1}\right)\left(\frac{2 \text{ mol Al}}{6 \text{ mol } e^-}\right)\left(\frac{26.98 \text{ g}}{1 \text{ mol Al}}\right) = 53.96 \text{ g Al}$

The $\boxed{\text{Zn electrode's}}$ mass changed more, since it gained 196.14 g in mass compared to the Al electrode's 53.96 g lost, if $6e^-$ were transferred between them, and this relationship is true regardless of how many e^-'s are transferred.

Part D

Largest difference = Au and Zn:

$E°_{cell} = E°_{cathode} - E°_{anode}$

$E°_{cell} = \underset{\text{Au}}{(1.50 \text{ V})} - \underset{\text{Zn}}{(-0.76 \text{ V})} = \boxed{+2.26 \text{ V}}$

Continue to Question 7.